Internet of Things

By

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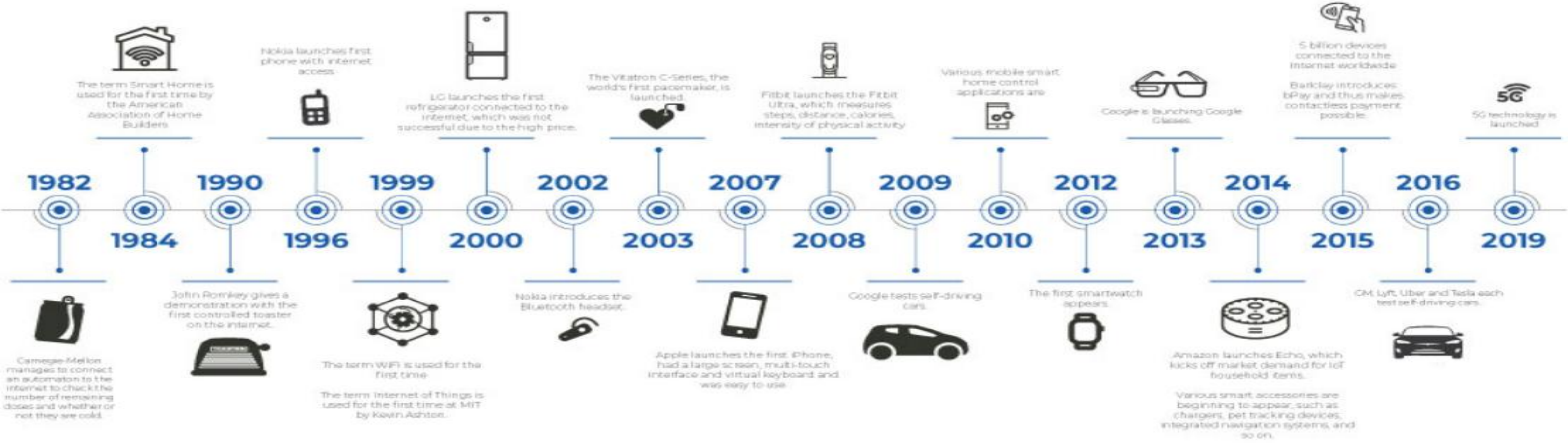
Outline

- Introduction of IoT
- History of the IoT
- IoT Potential
- Application and Evolution of IoT
- IoT concepts Technology
- IoT ecosystem
- IoT for global and Mass scaling
- References

History of IoT

The term IoT can most likely be attributed to Kevin Ashton in 1997 with his work at Proctor and Gamble using RFID tags to manage supply chains. The work brought him to MIT in 1999 where he and a group of like-minded individuals started the Auto-ID center research consortium. Since then, IoT has taken off from simple RFID tags to an ecosystem and industry that by 2020 will cannibalize, create, or displace five trillion out of one hundred trillion global GDP dollars, or 6% of the world GDP



History of the IoT

Year	Device	Reference
1973	Mario W. Cardullo receives the patent for first RFID tag	US Patent US 3713148 A
1982	Carnegie Mellon internet-connected soda machine	https://www.cs.cmu.edu/~coke/history_long.txt
1989	Internet-connected toaster at Interop '89	IEEE Consumer Electronics Magazine (Volume: 6, Issue: 1, Jan. 2017)
1991	HP introduces HP LaserJet IIISi: first Ethernet-connected network printer	http://hpmuseum.net/display_item.php?hw=350
1993	Internet-connected coffee pot at University of Cambridge (first internet-connected camera)	https://www.cl.cam.ac.uk/coffee/qsf/coffee.html
1996	General Motors OnStar (2001 remote diagnostics)	https://en.wikipedia.org/wiki/OnStar
1998	Bluetooth SIG formed	https://www.bluetooth.com/about-us/our-history
1999	LG Internet Digital DIOS refrigerator	https://www.telecompaper.com/news/lg-unveils-internetready-refrigerator--221266

History of the IoT

2000	First instances of *Cooltown* concept of pervasive computing everywhere: HP Labs, a system of computing and communication technologies that, combined, create a *web-connected experience for people, places, and objects*	https://www.youtube.com/watch?v=U2AkkuIVV-I
2001	First Bluetooth product launched: KDDI Bluetooth-enabled mobile phone	http://edition.cnn.com/2001/BUSINESS/asia/04/17/tokyo.kddibluetooth/index.html
2005	United Nation's International Telecommunications Union report predicting the rise of IoT for the first time	http://www.itu.int/osg/spu/publications/internetofthings/InternetofThings_summary.pdf
2008	IPSO Alliance formed to promote IP on objects, first IoT-focused alliance	https://www.ipso-alliance.org
2010	The concept of Smart Lighting formed after success in developing solid-state LED light bulbs	https://www.bu.edu/smartlighting/files/2010/01/BobK.pdf
2014	Apple creates iBeacon protocol for beacons	https://support.apple.com/en-us/HT202880

History of the IoT

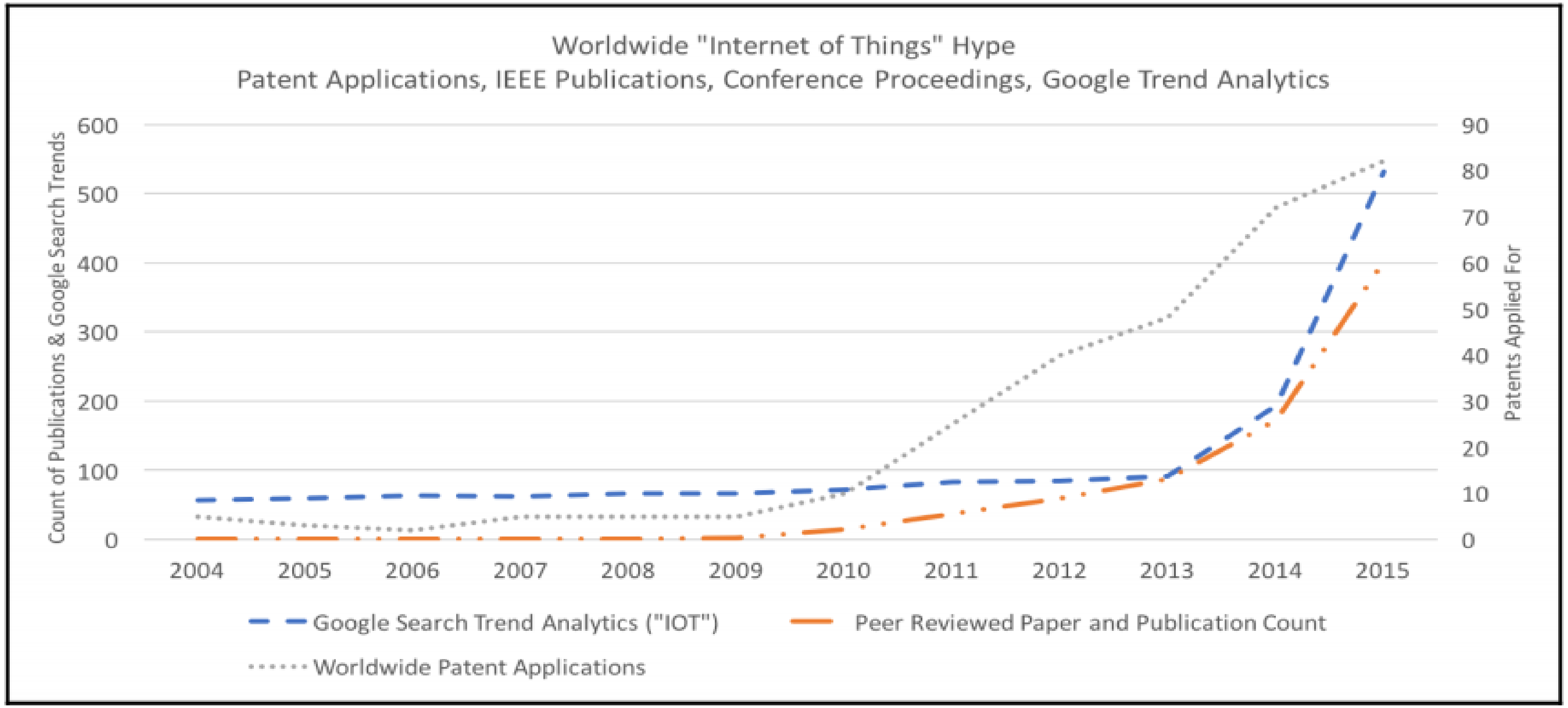


Fig. Analysis of keyword searches for IoT, patents, and technical publications

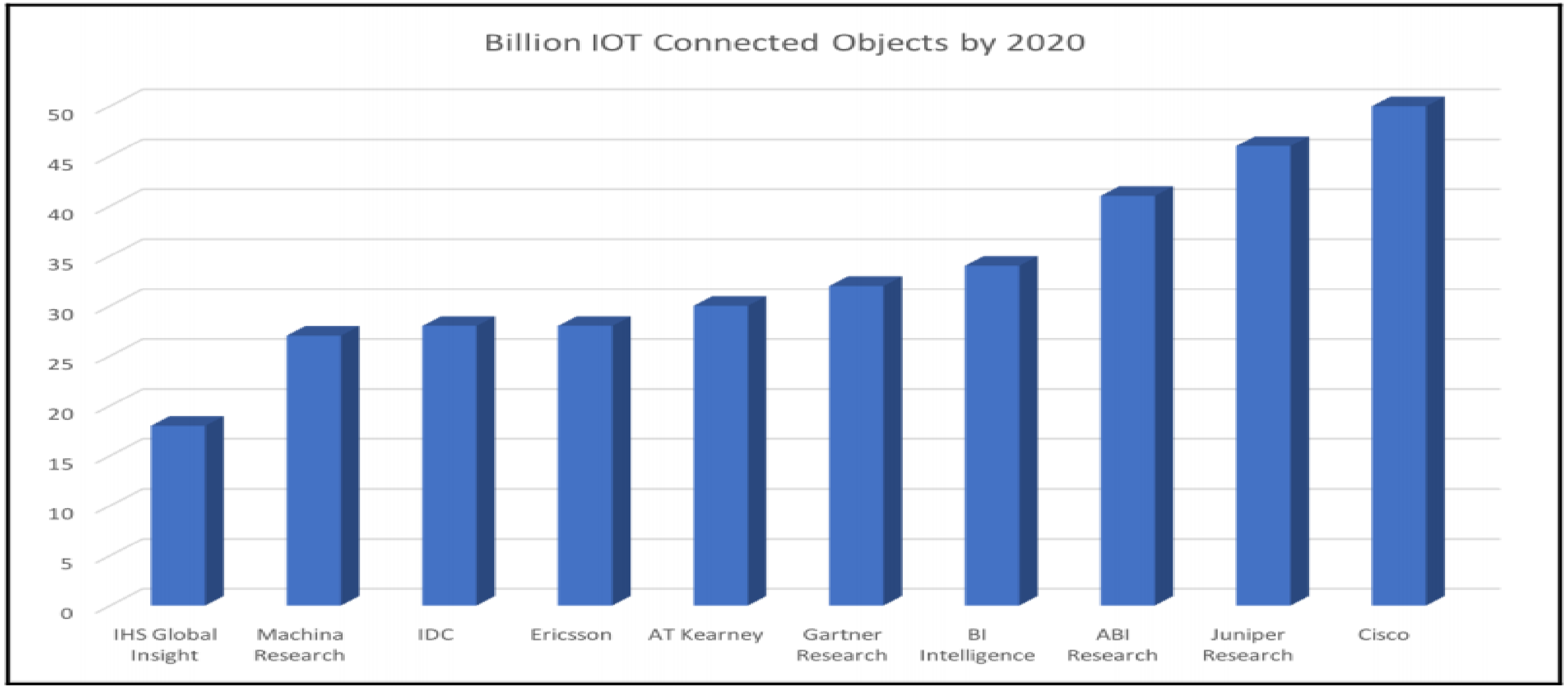
IoT Potential

- The IoT will touch nearly every segment in industrial, enterprise, health, and consumer products. It is important to understand the impact, as well as why these disparate industries will be forced to change in the way they build products and provide services.
- Perhaps your role as an architect forces you to focus on one particular segment; however, it is helpful to understand the overlap with other use cases.

IoT Potential

- ARM recently conducted a study and forecast that by 2035 one trillion connected devices will be operational. By all accounts, the projects growth rate in the near term is about 20 percent year over year.

IoT Potential



Different analyst and industry claims on the number of connected things

IoT design thinking

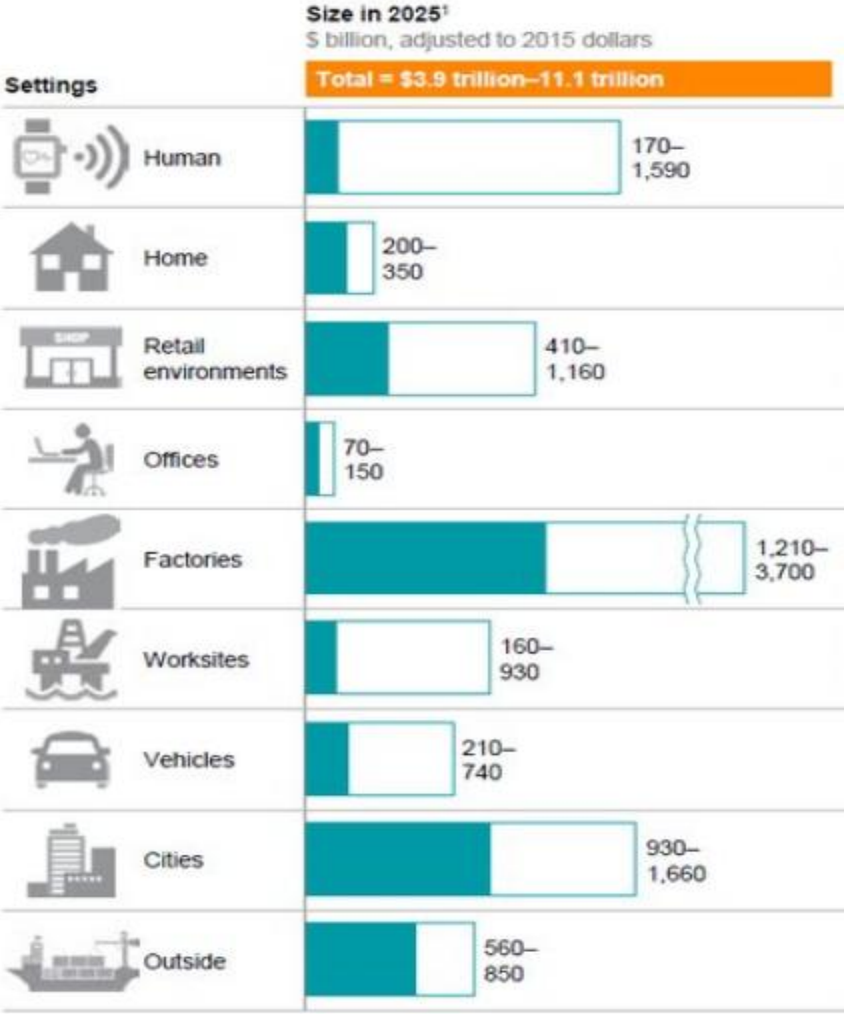
- IoT systems are not a fire-and-forget type of design. A customer expects several things from jumping on the IoT train.
- First, there must be a positive reward. That is dependent on your business, and your customer's intent. From authors experience, a 5x gain is the target and has worked well for the introduction of new technologies to pre-existing industries.

IoT design thinking

- Second, IoT design is, by nature, a plurality of devices. The value of IoT is not a single device or a single location broadcasting data to a server. It's a set of things broadcasting information and understanding the value the information in aggregate is trying to tell you. Whatever is designed must scale or will scale, therefore that needs attention in upfront design.

The impact of IoT

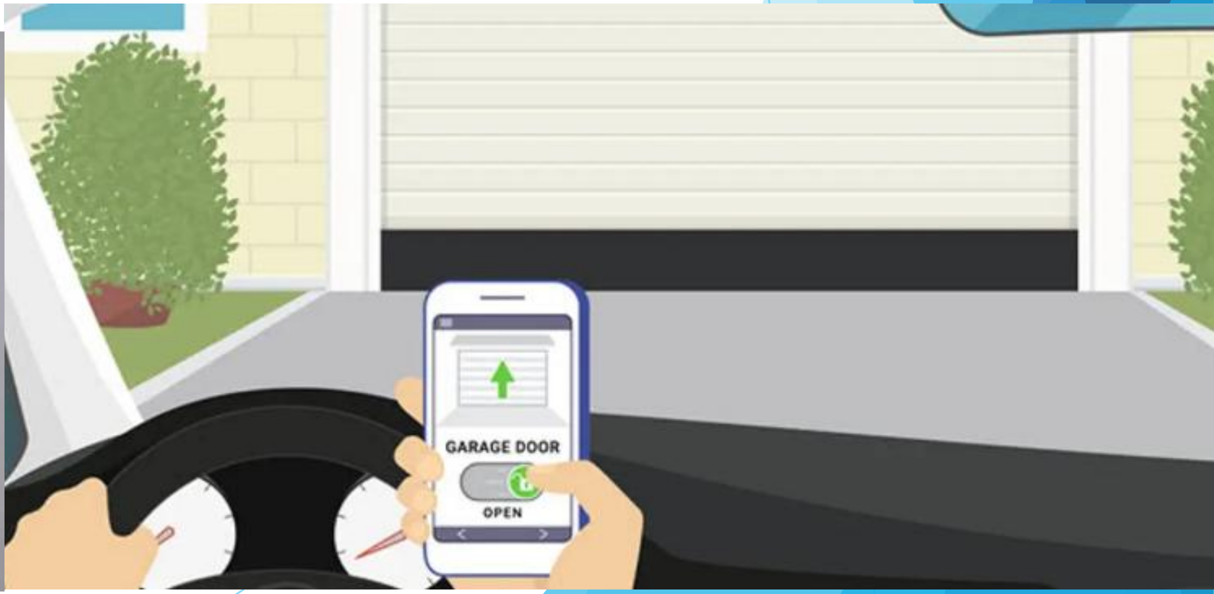
- New revenue streams (green energy solutions)
- Reducing costs (In-home patient healthcare)
- Reducing time to market (factory automation)
- Improving supply chain logistics (asset tracking)
- Reducing production loss (theft, spoilage of perishable)
- Increasing productivity (Machine learning and data analytics)
- Cannibalization (Nest replacing traditional thermostats)



If it is simply a new gadget, there will be a limited market scope. Only when the foreseeable benefit outweighs the cost will an industry thrive. The target used should be a 5x improvement over a traditional technology.

Consumer IoT use cases

- Smart home gadgetry: smart irrigation, smart garage doors, smart locks, smart lights, smart thermostats and smart security.
- Wearables: Health and movement trackers, smart clothing's/wearables.
- Pets: Pet location systems, smart dog doors.

Retail IoT use case

- Targeted advertising, such as locating known or potential customers by proximity and providing sales information.
- Beaconing, such as proximity sensing customers, traffic patterns, and inter-arrival times as marketing analytics.
- Asset tracking, such as inventory control, loss control, and supply chain optimizations.
- Cold storage monitoring, such as analyze cold storage of perishable inventory. Apply predictive analytics to food supply.

Retail IoT use case

- Insurance tracking of assets.
- Insurance risk measurement of drivers.
- Digital signage within retail, hospitality, or citywide.
- Beaconing systems within entertainment venues, conferences, concerts, amusement parks, and museums.

Healthcare

- The IoT is poised to allow for remote and flexible monitoring of patients wherever they may be. Advanced analytics and machine learning tools will observe patients in order to diagnose illness and prescribe treatments. Such systems will also be the watchdogs in the event of needed life-critical care.

  Currently, there are about 500 million wearable health monitors, with double digit growth in the years to come.

Healthcare IoT use cases

- In-home patient care
- Learning models of predictive and preventative healthcare
- Dementia and elderly care and tracking
- Hospital equipment and supply asset tracking
- Pharmaceutical tracking and security
- Remote field medicine
- Drug research
- Patient fall indicators

Transportation and logistics

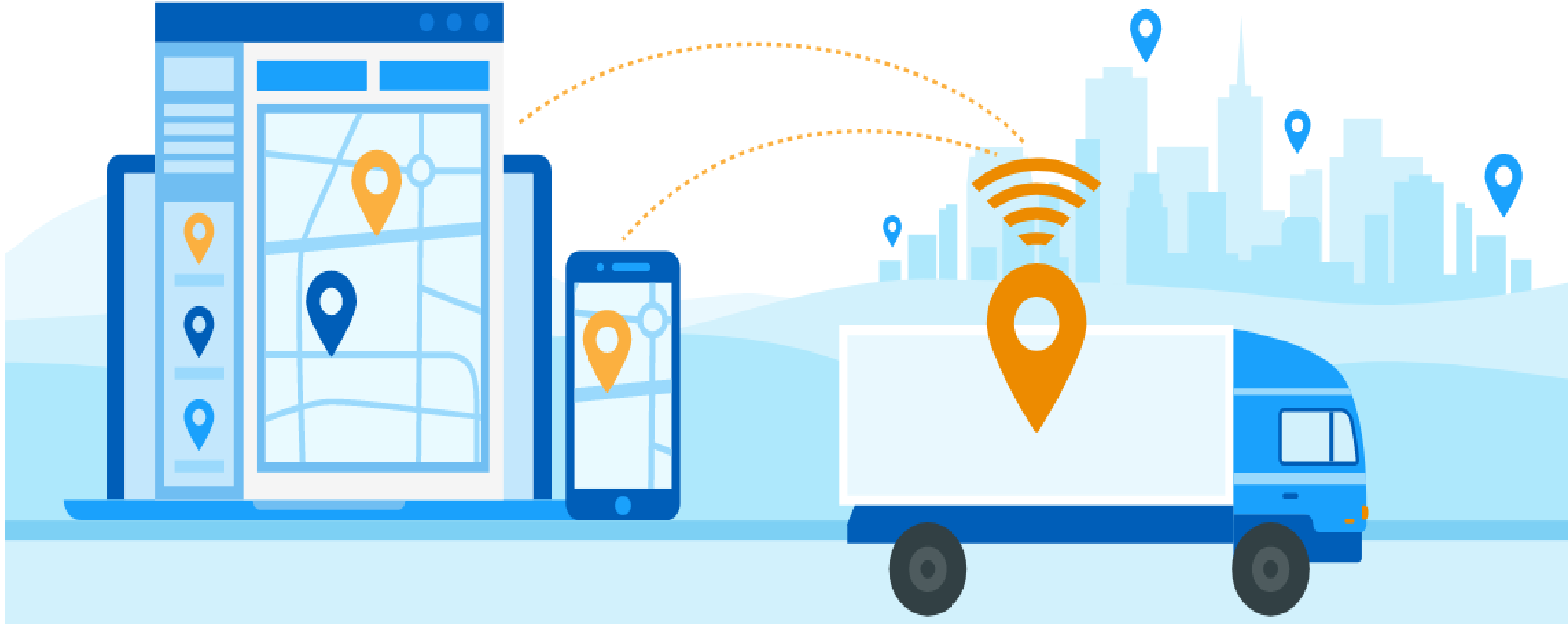
This is also the area of connected vehicles that communicate to offer assistance to the driver, or preventative maintenance on behalf of the driver. Right now, an average vehicle purchased new off a lot will have about 100 sensors. That number will double as vehicle-to-vehicle communication, vehicle-to-road communication, and automated driving become must-have features for safety or comfort.

Transportation and logistics

- This has important roles beyond consumer vehicles, and extends to rail lines and shipping fleets that cannot absorb any downtime. We will also see service trucks that can track assets within a service vehicle. Some of the use cases can be very simple, but also very costly, such as monitoring the location of service vehicles in the delivery of stock. Systems are needed to automatically route trucks and service personnel to locations based on demand versus routine.

Transportation and logistics IoT use cases

- Fleet tracking and location awareness
- Railcar identification and tracking
- Asset and package tracking within fleets
- Preventative maintenance of vehicles on the road

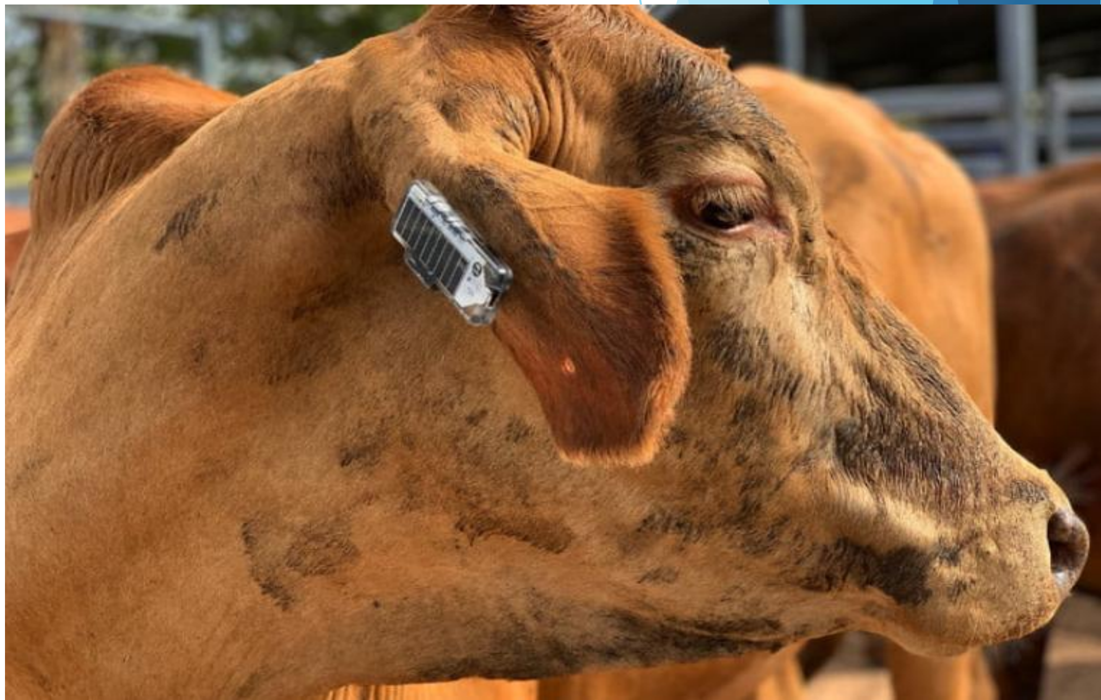


Agricultural and environmental

Farming and environmental IoT includes elements of livestock health, land and soil analysis, micro-climate predictions, efficient water usage, and even disaster predictions in the case of geological and weather-related disasters. Even as the world population growth slows down, world economies are becoming more affluent. Hunger and starvation crises are rare. The demand for food production is set to double by 2035. Significant efficiencies in agriculture can be achieved through IoT. Using smart lighting to adjust the spectrum frequency based on poultry age can increase growth rates and decrease mortality rates based on stress on chicken farms. Additionally, smart lighting systems could save $1 billion annually.

Agricultural and environmental

- Other uses include detecting livestock health based on sensor movement and positioning. A cattle farm could find animals with the propensity of sickness before a bacterial or viral infection were to spread. Edge analysis systems could find, locate, and isolate heads of cattle in real time, using data analytics or machine learning approaches.



Agricultural and environmental IoT use cases

- Smart irrigation and fertilization techniques to improve yield
- Smart lighting in nesting or poultry farming to improve yield
- Livestock health and asset tracking
- Preventative maintenance on remote farming equipment via manufacturer
- Drones-based land surveys
- Farm-to-market supply chain efficiencies with asset tracking
- Robotic farming
- Volcanic and fault line monitoring for predictive disasters

Energy

The energy segment includes the monitoring of energy production at source to and through the usage energy at the client. A significant amount of research and development has focused on consumer and commercial energy monitors such as smart electric meters that communicate over low-power and long-range protocols to reveal real-time energy usage.

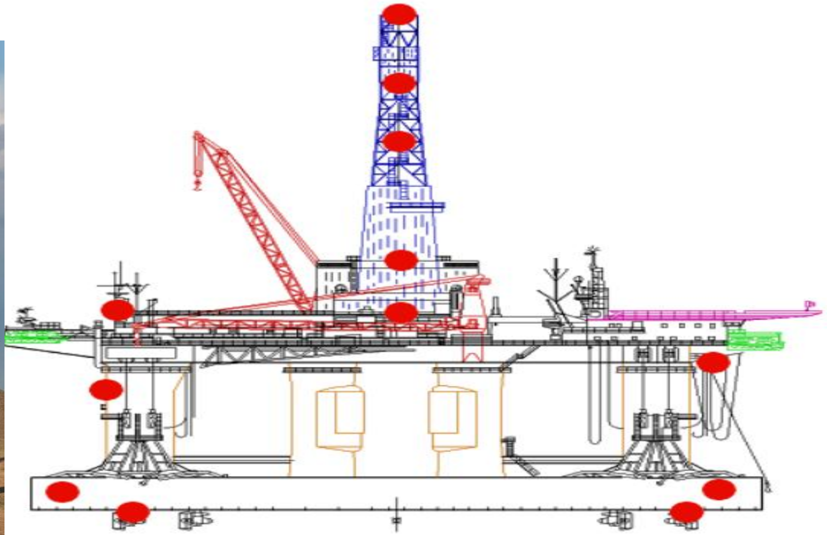
Energy

- Many energy production facilities are in remote or hostile environments such as desert regions for solar arrays, steep hillsides for wind farms, and hazardous facilities for nuclear reactors. Additionally, data may need real-time or near real-time response for critical response to energy production control systems

Energy IoT use cases

The following are some of the use cases for energy IoT:

- Oil rig analysis of thousands of sensors and data points for efficiency gains
- Remote solar panel monitoring and maintenance.
- Hazardous analysis of nuclear facilities
- Smart electric meters in citywide deployment to monitor energy usage and demand.
- Real-time blade adjustments as a function of weather on remote wind turbines



Smart City

Smart cities are one of the fastest growing segments, and show substantial cost/benefit ratios especially when we consider tax revenues. Smart cities also touch citizens' lives through safety, security, and ease of use.

For example, several cities

Such as Barcelona are fully connected and monitor trash containers and bins for pickup based on the current capacity, but also the time since the last pickup. This improves the trash collection efficiency allowing the city to use fewer resources and tax revenue in transporting waste, but also eliminates potential smells and odors of rotting organic material.

Smart city IoT use cases

- Pollution control and regulatory analysis through environmental sensing
- Micro climate weather predictions using city wide sensor networks
- Efficiency gains and improved costs through waste management service on demand
- Improved traffic flow and fuel economy through smart traffic light control and patterning
- Energy efficiency of city lighting on demand

Smart city IoT use cases

- Smart snow plowing based on real-time road demand, weather conditions, and nearby plows
- Smart irrigation of parks and public spaces, depending on weather and current usage
- Smart cameras to watch for crime and real-time automated AMBER Alerts (America's Missing: Broadcast Emergency Response)
- Smart parking lots to automatically find best space parking on demand
- Bridge, street, and infrastructure wear and usage monitors to improve longevity and service

Government and military

City, state, and federal governments, as well as the military, have a keen interesting in IoT deployments.

California's executive order B-30-15 which states that by 2030 greenhouse gas emissions affecting global warming will be at levels 40 percent below 1990 levels. To achieve aggressive targets like this, environmental monitors, energy sensing systems, and machine intelligence will need to come into play to alter energy patterns on demand while still keeping the California economy breathing.

Government and military IoT use cases

- Terror threat analysis through IoT device pattern analysis
- Swarm sensors through drones
- Sensor bombs deployed on the battlefield to form sensor networks to monitor threats
- Government asset tracking systems
- Real-time military personal tracking and location services
- Synthetic sensors to monitor hostile environments
- Water level monitoring to measure dam and flood containment

IoT Architecture and Core IoT modules (Cheapter-2)

- Architect's role in building a scalable, secure, and enterprise IoT architecture.
- To do that, an architect must be able to speak to the value the design brings to a customer.
- The architect must also play multiple engineering and product roles in balancing different design choices.

IoT ecosystem

Nearly every major technology company is investing or has invested heavily in IoT space.

Sensors: Embedded systems, real-time operating systems, energy-harvesting sources, Micro-Electro-Mechanical Systems(MEMs).

Sensor communication systems: Wireless personal area networks reach from 0 cm to 100 m. Low-speed and low-power communication channels, often non-IP based have a place in sensor communication.

IoT ecosystem

Local area networks: Typically, IP-based communication systems such as 802.11 Wi-Fi used for fast radio communication, often in peer-to-peer or star topologies.

Aggregators, routers, gateways: Embedded systems providers, cheapest vendors (processors, DRAM, and storage), module vendors, passive component manufacturers, thin client manufacturers, cellular and wireless radio manufacturers, middleware providers, fog framework providers, edge analytics packages, edge security providers, certificate management systems.

IoT ecosystem

WAN: Cellular network providers, satellite network providers, Low-Power Wide-Area Network(LPWAN) providers. Typically using internet transport protocols targeted for IoT and constrained devices like MQTT, CoAP, and even HTTP.

Cloud: Infrastructure as a service provider, platform as a service provider, database manufacturers, streaming and batch processing manufacturers, data analytics packages, software as a service provider, data lake providers, Software Defined Networking/Software-Defined Perimeter providers, and machine learning services.

IoT ecosystem

Data analytics: As the information propagates to the cloud. Dealing with volumes data and extracting value is the job of complex event processing, data analytics, and machine learning techniques.

Security: Tying the entire architecture together is security. Security will touch every component from physical sensors to the CPU and digital hardware, to the radio communication systems, to the communication protocols themselves. Each level needs to ensure security, authenticity, and integrity. There cannot be the weak link in a chain, as the IoT will form the largest attack surface on earth.

IoT vs M2M

M2M: It is a general concept involving an autonomous device communicating directly to another autonomous device. Autonomous refers to the ability of the node to instantiate and communicate information with another node without human intervention. The form of communication is left open to the application. It may very well be the case that an M2M device uses no inherent services or topologies for communication. This leaves out typical internet appliances used regularly for cloud services and storage.

IoT vs M2M

- **IoT:** IoT systems may incorporate some M2M nodes (such as a Bluetooth mesh using non-IP communication), but aggregates data at an edge router or gateway. An edge appliance like a gateway or router serves as the entry point onto the internet. Alternatively, some sensors with more substantial computing power can push the internet networking layers onto the sensor itself. Regardless of where the internet on-ramp exists, the fact that it has a method of tying into the internet fabric is what defines IoT.

The value of a network and Metcalfe's and Beckstrom's law

Robert Metcalfe in 1980 formulated the concept that the value of any network is proportional to the square of connected users of a system. In the case of IoT, users may mean sensors or edge devices. Metcalfe's law is represented as:

$$V \propto N^2$$

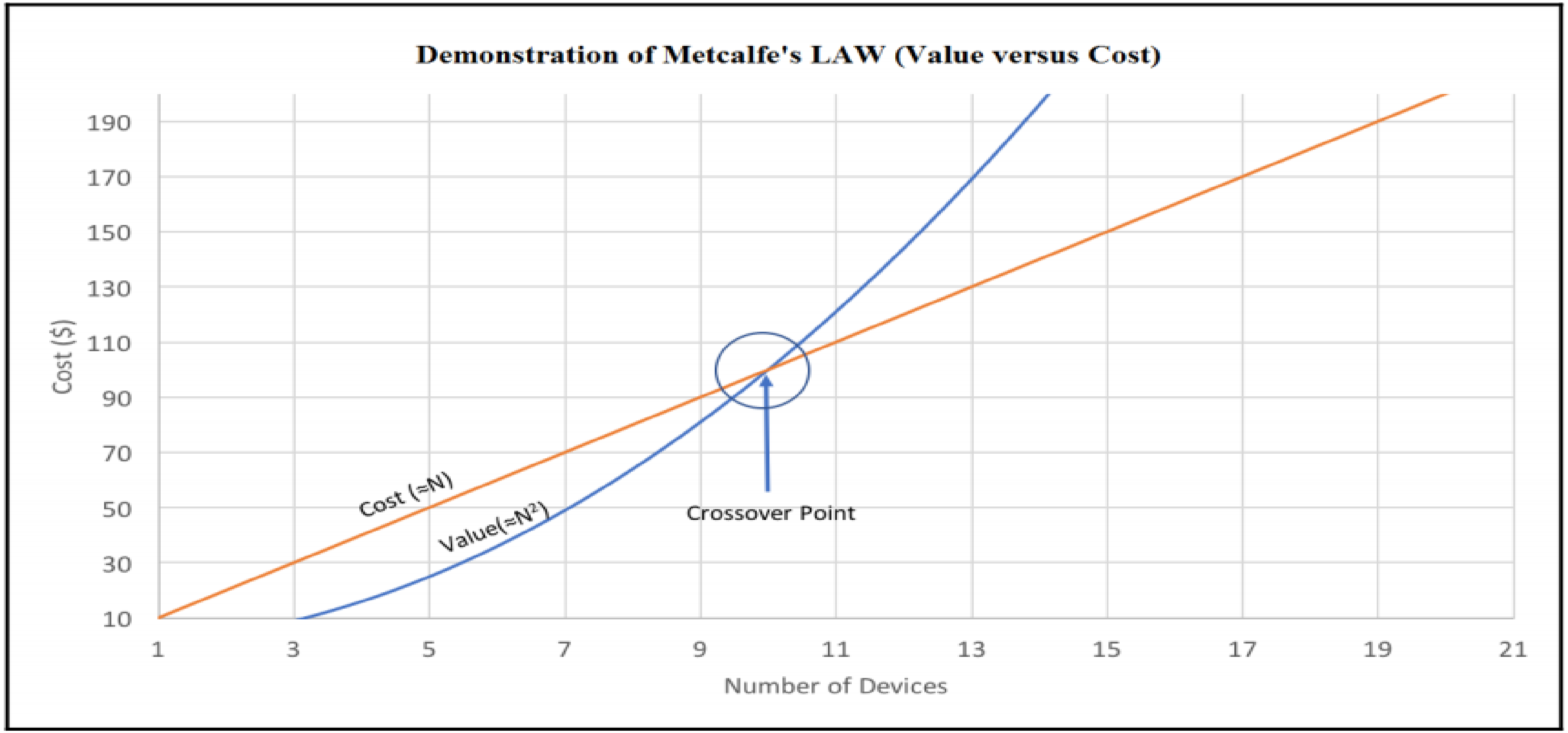
Where:

V= Value of the network

N= Number of nodes within the network

A graphical model helps to understand the interpretation as well as the Crossover Point, where a positive return on investment (ROI) can be expected:

IoT Architecture and Core IoT Modules



Metcalfe's Law. Value of a network is represented as proportional to N (squre). The cost of each node is represented as kN where k is an arbitrary constant. In this case, k represents a constant of $10 per IoT edge sensor. The key takeaway is the crossover point occurs rapidly due to the expansion of value and indicates when this IoT deployment achieves a positive ROI.

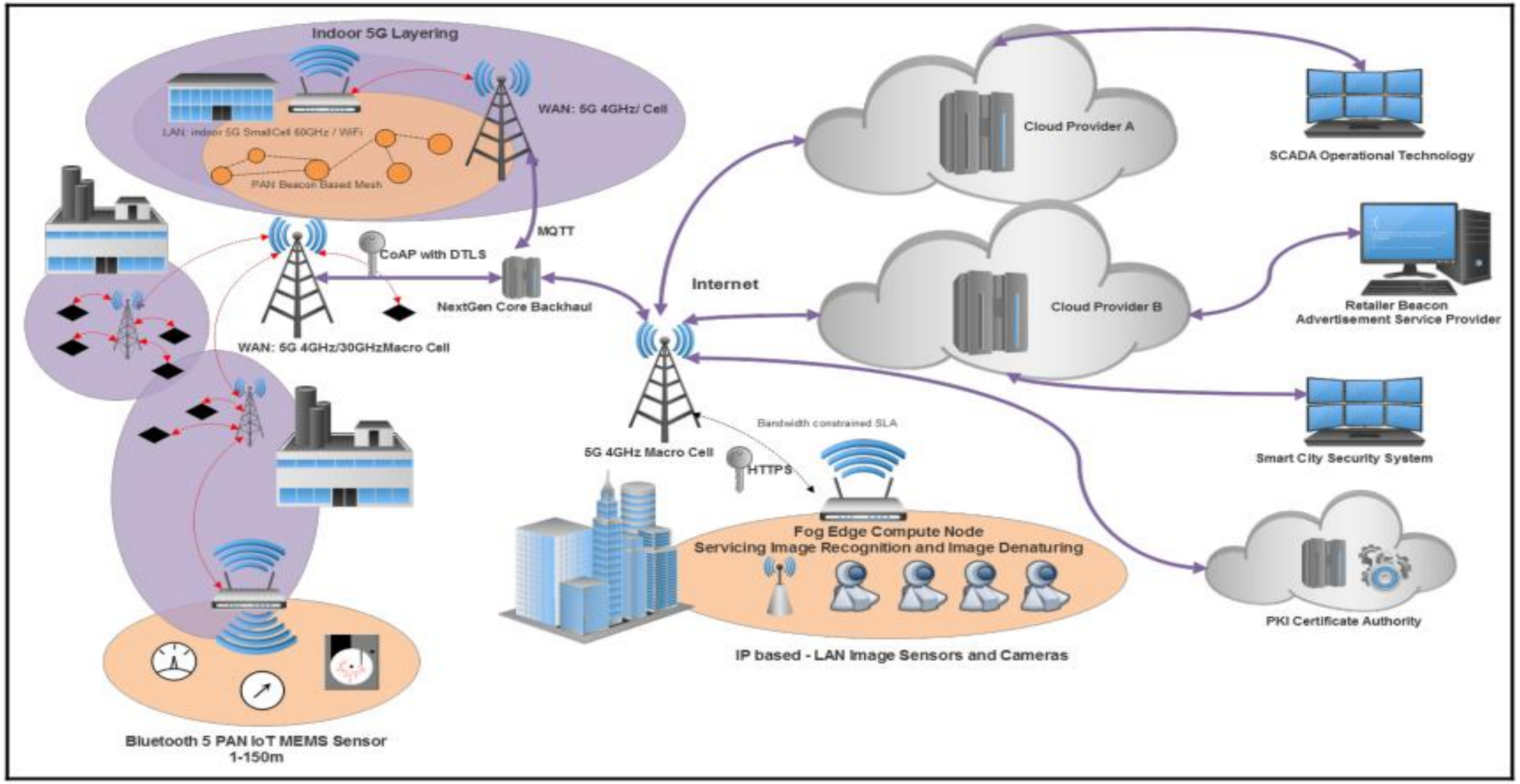
IoT architecture

The architect needs to consider interference effects in the LAN and WAN—how will the data get off the edge and on the internet? The architect needs to consider resiliency and how costly the loss of data is. Should resiliency be managed within the lower layers of the stack, or in the protocol itself?

The architect must also make choices of internet protocols such as MQTT versus CoAP and AMQP, and how that will work if he or she decides to migrate to another cloud vendor.

This opens up the notion of fog computing to process data close to its source to solve latency problems, but more importantly to reduce bandwidth and costs moving data over WANs and clouds

IoT architecture



There are now over 1.5 million different combinations of architectures to choose from:

Fig: The full spectrum of various levels of IoT architecture from the sensor to cloud and back.

IoT for global and Mass scaling

- Sensing and power: (part-1)

Sometimes this involves considerable data being generated from a single sensor, such as auditory sensing for preventative maintenance of machinery. Other times, it's a single bit of data indicating vital health data from a patient.

Collections of billions of small sensors will still require a massive amount of energy in whole to power.

IoT for global and Mass scaling

- Data communication (part-2)

The IoT wouldn't exist without significant technologies to move data from the remotest and most hostile environment to the largest data centers at Google, Amazon, Microsoft, and IBM.

The starting point for IoT is about connectivity. A successful architect will understand the constraints of internetworking from a sensor to a WAN and back again.

IoT for global and Mass scaling

- Internet routing and protocols (part-3)

To bridge data from sensors to the internet, two technologies are needed: **gateway routers** and **supporting IP-based protocols** designed for efficiency. The role of the router is especially important in securing, managing, and steering data. Edge routers orchestrate and monitor underlying mesh networks and balance and level data quality. The privatization and security of data is also critical.

IoT for global and Mass scaling

- Fog and edge compute, analytics, and machine learning (Part 4)

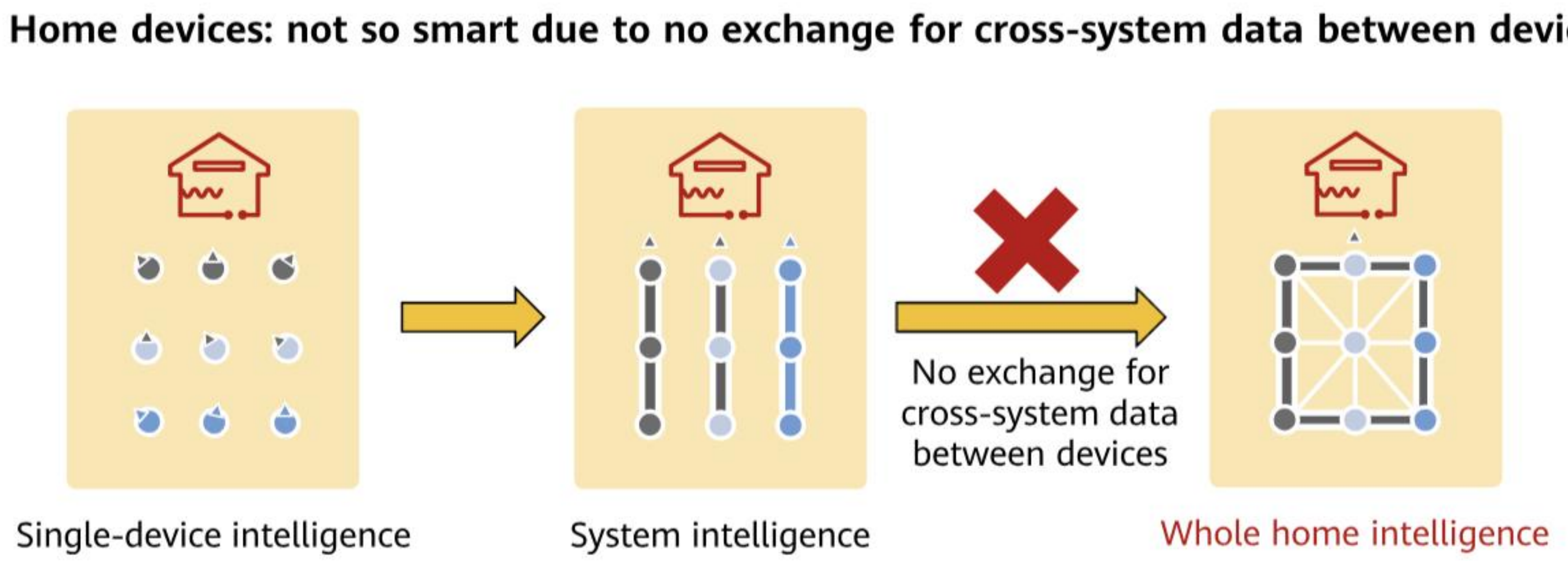
At this point, we must consider the aspects of cloud architectures such as SaaS, IaaS, and PaaS systems. An architect needs to understand the data flow and typical design of cloud services (what they are and how they are used). We use OpenStack as a model of cloud design and explore the various components from ingestion engines, to data lakes, to analytics engines. Understanding the constraints of cloud architectures is also important to make a good judgment on how a system will deploy and scale. An architect must also understand how latency can affect an IoT system. Alternatively, not everything belongs in the cloud. There is a measurable cost in moving all IoT data to a cloud versus processing it at the edge.

IoT for global and Mass scaling

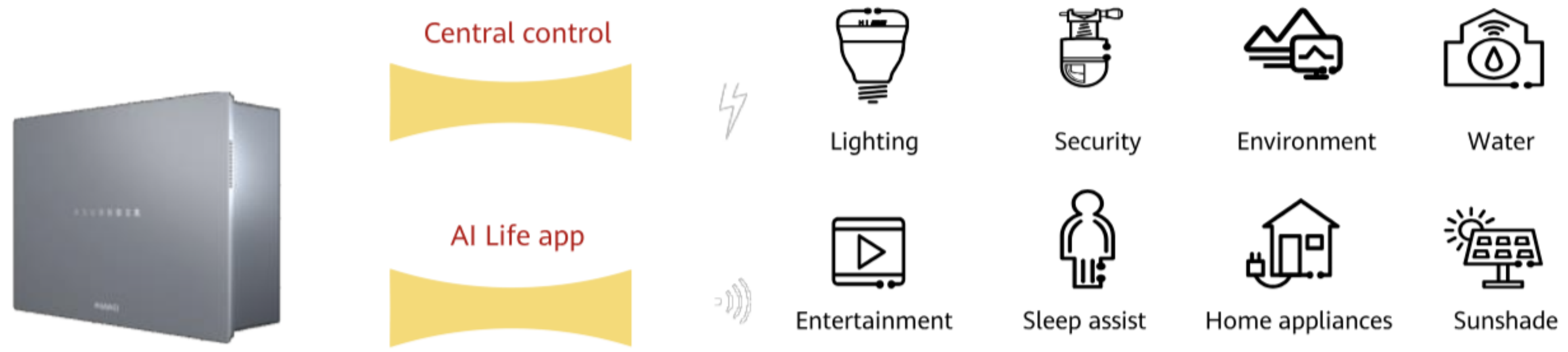
- Threat and security in IoT (part-5)

In many cases, IoT systems will not be secured in a home, or in a company. They will be in the public, in very remote areas, in moving vehicles, or even inside a person. The IoT represents the single biggest attack surface for any type of cyber attack. We have seen countless academic hacks, well-organized cyber assaults, and even nation-state security breaches with IoT devices being the target.

Common issue: Different Connection standards Bring Fragmented Interaction experience



All in one smart home solution



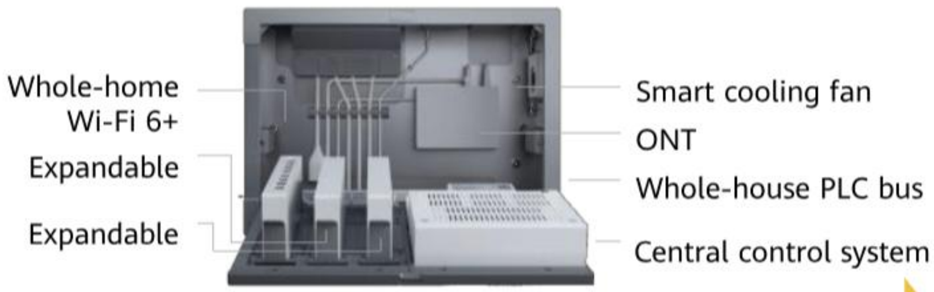
1 host: computing center

2 solutions: 2 interaction modes

N subsystems: HarmonyOS Connect ecosystem

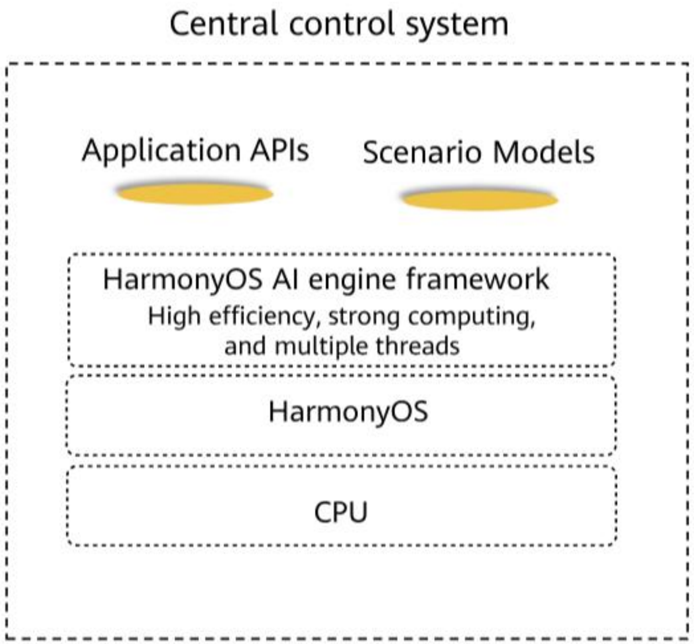
Core: Harmony-powered smart host + two interaction modes + diverse HarmonyOS Connect ecosystem products

All in one smart home solution

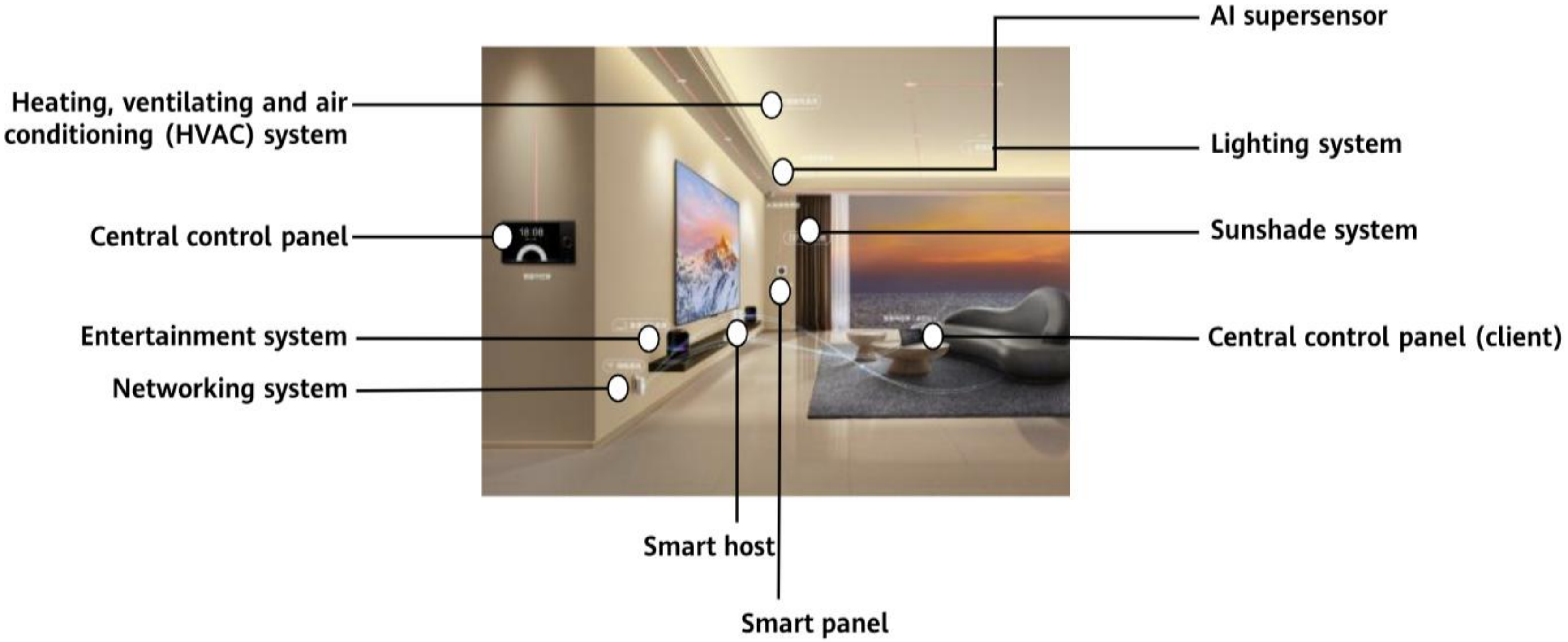


Smart hosts running on HarmonyOS

- Steady function upgrade
- Dynamic prediction based on multiple conditions to build immersive and personalized AI



All in one smart home solution



Common challenges in automative industry

Security

- Delayed detection of vehicle faults
- Impact of faulty vehicles on safety of other vehicles
- Influence of weather changes on driving

New value

- Fixed insurance rates
- Private use of company cars
- Fleet management
- ETC, parking fee recharge, etc.

Efficiency

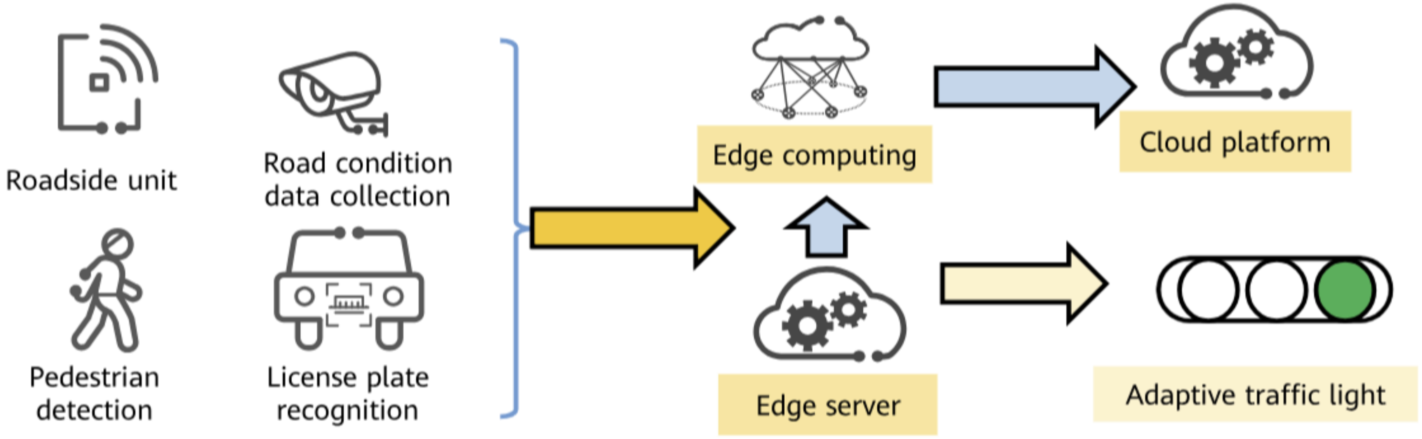
- Traffic citation for speeding
- Traffic block due to road repair
- No smooth music playing
- No smart control on traffic lights

Internet of Vehicles (IoV)

- Internet of Vehicles (IoV) means that **vehicle-mounted devices** use **wireless communication technologies** to make full use of all dynamic vehicle information on the **information network platform** and **provide various functions and services** during vehicle running.
- IoV has the **following characteristics**: Provides assurance for a distance between vehicles to reduce a probability of vehicle collision accidents. Provides real-time navigation for drivers and communicates with other drivers and network systems to improve traffic efficiency.

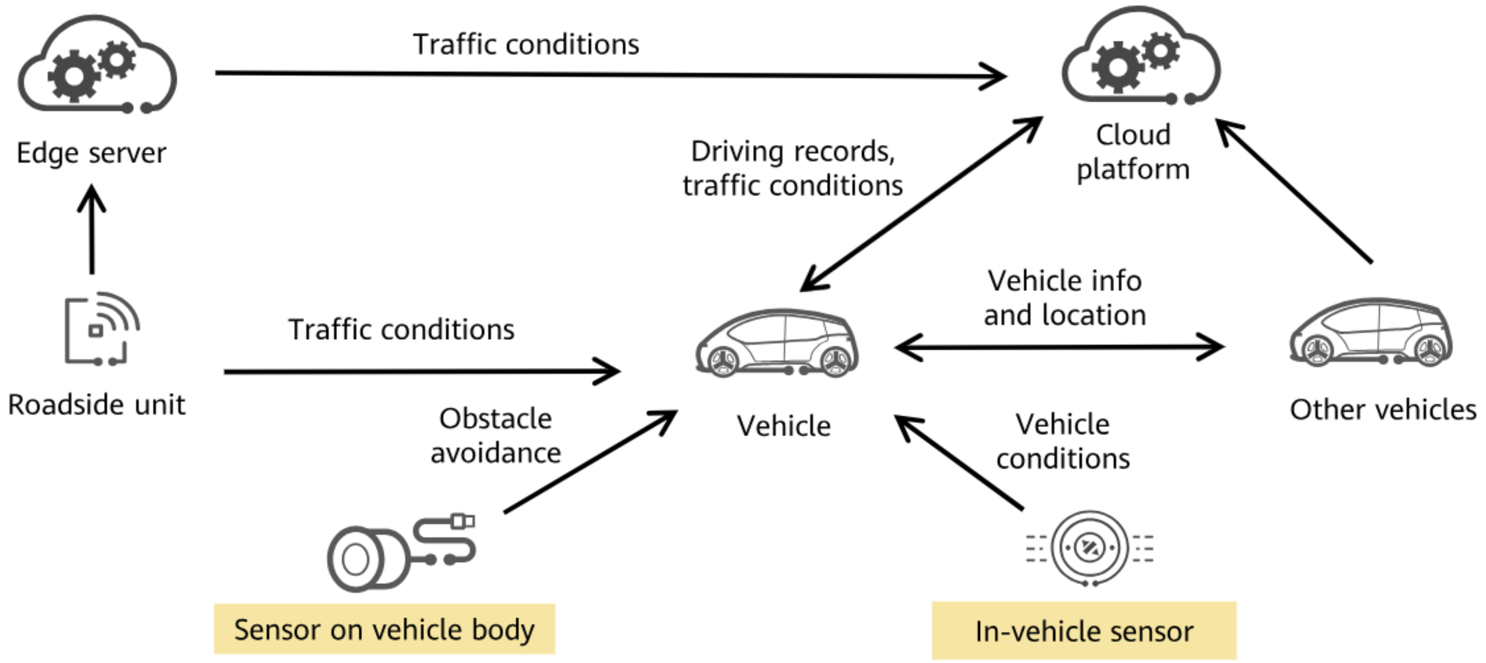
Building smart solution based on the Internet of Vehicles (IoV) Solution

- Traffic conditions change at all hours of the day, and traffic lights are key to controlling traffic.
- Traditional solution: Traffic lights turn on and off in a specified sequence, which cannot change according to road conditions.
- Smart intersections based on IoV: Road condition data is collected to the edge to adjust the traffic lights. The cloud platform stores such data for big data analytics, providing data basis for future urban planning.



Vehicle to X

- V2X is closely related to smart transportation and intelligent roads.



Vehicle to X

- Vehicle-To-Network (V2N) is one of the most widely used connections, which allows vehicles to connect to cloud servers through a mobile network, thereby implementing application functions such as navigation, entertainment, and anti-theft. A common application scenario is that a vehicle-mounted system connects to the Internet through a mobile phone hotspot to obtain real-time navigation information or play online music. Currently, many vehicle-mounted systems provide card ports, which can be directly inserted with data cards for the connection to the Internet.
- Vehicle-To-Vehicle (V2V) aims to implement collision warning and congestion avoidance functions through information exchange between vehicles. For example, before entering the corner of a winding road, a veteran driver would honk to warn the vehicles in the blind zone. With the V2V technology, drivers can communicate with each other without using speakers. In addition, information such as speed and direction can also be transmitted among them.

Vehicle to X

- Vehicle-To-Infrastructure (V2I) enables vehicles to exchange data with roads and roadside infrastructure, such as traffic lights, road signs, and even roadblocks. In addition, vehicles can collect information about the surrounding environment. This kind of connection seems to be not helpful to drivers, but plays an obvious role in the automated driving field because the vehicle can find an obstacle in the front by using this connection.
- V2P (Vehicle-To-Pedestrian) is similar to V2V and aims to prevent collisions through information exchange. In addition to cameras and sensors, information interconnection is also the most effective way to detect pedestrians. For example, a terminal used by a pedestrian, such as a mobile phone, a tablet, or a wearable device, can implement interconnection between a person and a vehicle. This helps to prevent staged accident fraud, or "pengci".
- As one of the IoV solutions, Vehicle to Everything (V2X) integrates digital technologies such as cloud, IoT, AI, and 5G to implement efficient collaboration among people, vehicles, roads, and clouds, build smart vehicles and roads, and

Digital Road infrastructure service

- DRIS implements digital information exchange among pedestrians, vehicles, roads, and networks, improves driving safety and traffic efficiency, and facilitates large-scale commercial use of autonomous driving.
- DRIS consists of **V2X Server in the cloud** and **V2X Edge at the edge**.
  - V2X Server provides digital road infrastructure services and edge-cloud synergy services such as data analysis and roadside computing unit management.
  - V2X Edge provides real-time service processing such as roadside sensor data collection, event identification, and communications.
- The goal of DRIS is to connect multiple roadside sensors for digital perception and provide information for efficient traffic.

IoV development history

Road information collection and communications

Improve the efficiency of certain roads with the traffic control and monitoring system for single points or certain areas.

Selective traffic control

Wired and wireless roadside communications

Telematics

Vehicle-to-network WAN communication implements online navigation, remote diagnosis and control, infotainment, and vehicle alarm through in-vehicle communication modules and cellular communication.

Online navigation/IVI

2G/3G/4G

ITS - BSA

V2V or V2I short-distance communication implements reminding or even controls the potential collision risks of vehicles, improving vehicle safety and traffic efficiency (basic application set).

Assisted driving

802.11p/LTE-V

ITS - ESA

Realize automatic control and automated driving without accidents for real convergence of pedestrians, vehicles, roads, and environments. It is the future ITS.

Autonomous driving

Future technologies

1990 — 2005 — 2015 — 2020 — 2025 — 2030

Phase 1: Entertainment and navigation functions

- Entertainment: In the past, in-vehicle infotainment (IVI) uses cassettes, CDs, and MP3 downloaded by using USB drives. Now, drivers can listen to music online through IoV.
- Navigation: Real-time and connected map navigation replaces the map navigation stored on SD cards.

Phase 2: Attached ecosystem and O2O of the IoV represented by OBD/T-Box

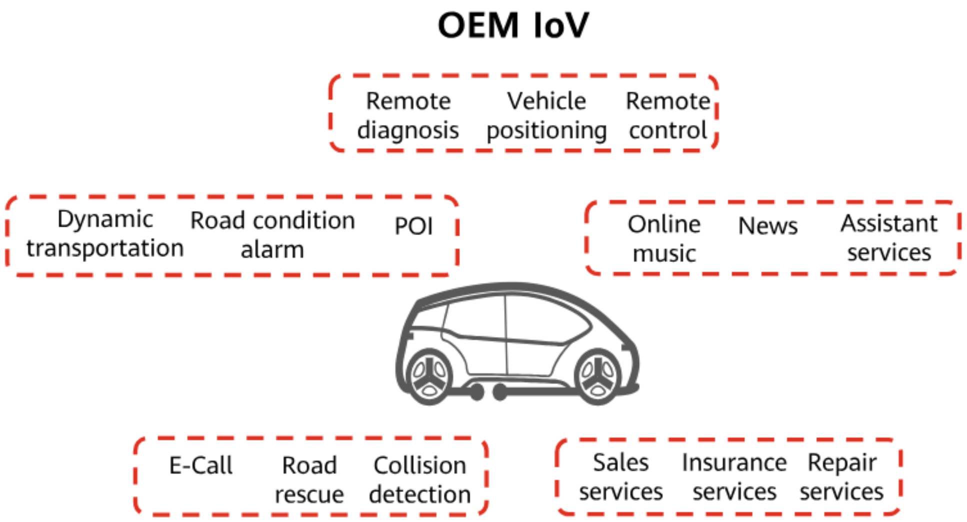
- Connect to the Electronic Control Unit (ECU) to obtain more vehicle information, such as mileage, fuel consumption, fault, and location, for subsequent services.
- Provide sufficient map information points, and even directly complete O2O consumption. This is not only an IoV problem, but also an O2O consumption entry.

Phase 3: Automated driving and auxiliary driving substitute for the existing HU

- Integrate the sensing capability of sensors, judgment capability of processors, and control capability of electric controllers.
- The core of automated driving is processor judgment. The hardware and software of this processor are highly artificial intelligence. Once the automated driving function is implemented, both the ECU and HU will be replaced.

Today's IoV

- IoV is evolving from IVI services to smart transportation. The OEM mode focuses on internal services of OEMs, vehicle data collection, and personal entertainment information service. The aftermarket mode focuses on industry applications, supplemented by personal IVI information service.



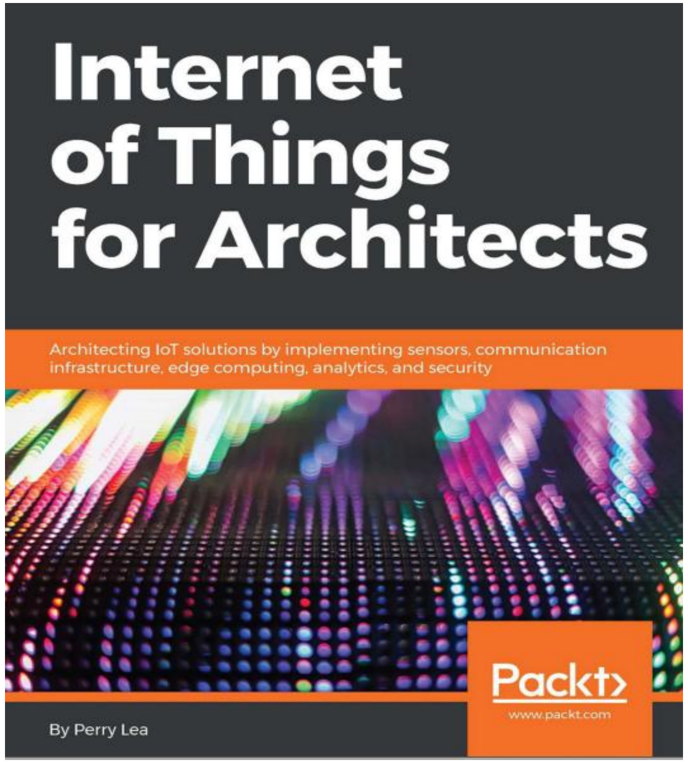
Aftermarket IoV

To B	UBI	Fleet management
To C	Smart rearview mirrors	In-vehicle Wi-Fi

Industrial IoT development plan across countries

	2014–2017	2020–2022 Industrial IoT	2025
Germany	✓ Implemented Industry 4.0.	✓ Complete manufacturing **communications standardization**.	✓ **Unify EU Industry 4.0 standards**.
China	✓ Released *China Manufacturing 2025* initiative.	✓ Establish a **standards system** for smart manufacturing communications equipment. ✓ Chinese industrial robots occupy 50% of the market. ✓ Large-scale use of industrial wireless networks with a bandwidth of **500 Mbit/s**	✓ Breakthroughs in **10** fields, such as automotive, healthcare, and energy ✓ Large-scale use of industrial wireless communications networks with a bandwidth of **2 Gbit/s**
US	✓ Released the *Revitalize American Manufacturing and Innovation Act.*	✓ Complete flexible production line assembly within 24 hours.	✓ Invest in 1.9 billion dollars to build **45** innovation organizations. ✓ Complete flexible production line assembly within 8 hours.
Japan	✓ Released new strategies for robots and IoT. ✓ Released the Industrial Value Chain Initiative (IVI).	✓ Complete **international standardization** of manufacturing robots. ✓ Large-scale commercial use of service robots.	✓ Improve manufacturing informatization level from 30% to **50%**.
South Korea	✓ Proposed *Manufacturing Industry Innovation 3.0.*	✓ Develop IoT/smart manufacturing, make **30% existing factories** intelligent, and develop 10,000 intelligent production lines.	✓ Invest 23 billion dollars in **13** industries, such as UAVs, smart vehicles, and healthcare. The export amount exceeds Japan's.

Reference book



Chapter 1 and chapter 2

References

- 1. Book - Internet of Things for Architects (chapter 1 and chapter 2)

  By Perry Lea

Question

Thank you